
Association Rule Mining - I

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Market Basket Analysis

Scenario: customers shopping at a supermarket

Transaction id	Items
1	Bread, Ham, Juice, Cheese, Salami, Lettuce
2	Rice, Dal, Coconut, Curry leaves, Coffee, Milk, Pickle
3	Milk, Biscuit, Bread, Salami, Fruit jam, Egg
4	Tea, Bread, Salami, Bacon, Ham, Sausage, Tomato
5	Rice, Egg, Pickle, Curry leaves, Coconut, Red chilly

- What can we infer from the above data?
- An association rule: {Bread, Salami} $\rightarrow$ {Ham}, with confidence $\sim 2/3$

Association Rule Mining

- Proposed by Agrawal et al in 1993.
- It is an important data mining model studied extensively by the database and data mining community.
- Assume all data are categorical.
- No good algorithm for numeric data.
- Initially used for **Market Basket Analysis** to find how items purchased by customers are related.

Bread → Milk [sup = 5%, conf = 100%]

Applications

- Information driven marketing
- Catalog design
- Store layout
- Customer segmentation based on buying patterns

- Several papers by Rakesh Agrawal and others in the 1990s
- Rakesh Agrawal and Ramakrishnan Srikant
Fast Algorithms for Mining Association Rules
The VLDB 1994

The Market-Basket Model

- A (large) set of *binary attributes*, called *items*
 $I = \{i_1, \dots, i_n\}$
e.g. milk, bread, the items sold at the market
- A *transaction* T consists of a (small) subset of I
e.g. the list of items (bill) bought by one customer at once
- The *database* D is a (large) set of transactions
 $D = \{T_1, \dots, T_N\}$

The Market-Basket Model

- **Goal:** mining associations between the *items*
 - The transactions or customers also may have associations, but here we are interested in such relations
- **Approach:** finding subset of items that are present together in transactions frequently
- An *itemset*: any subset X of I

Transaction data: supermarket data

- Market basket transactions:

T1: {bread, cheese, milk}

T2: {apple, eggs, salt, yogurt}

...

...

Tn: {biscuit, eggs, milk}

- Concepts:

- *An item*: an item/article in a basket
- *I*: the set of all items sold in the store
- *A transaction*: items purchased in a basket; it may have TID (transaction ID)
- *A transactional dataset*: A set of transactions

Support of an Itemset

- Let X be an itemset
- Support count $\sigma(X) = \#$ of transactions containing all items of X
- $\text{support}(X) = \text{fraction of transactions containing all items of } X$

$\text{support}(\{\text{Bread}, \text{Salami}\}) = 0.6$

$\text{support}(\{\text{Rice}, \text{Pickle}, \text{Coconut}\}) = 0.4$

T-ID	Items
1	Bread, Ham, Juice, Cheese, Salami, Lettuce
2	Rice, Dal, Coconut, Curry leaves, Coffee, Milk, Pickle
3	Milk, Biscuit, Bread, Salami, Fruit jam, Egg
4	Tea, Bread, Salami, Bacon, Ham, Sausage, Tomato
5	Rice, Egg, Pickle, Curry leaves, Coconut, Red chilly

- Makes sense (statistically significant) only when
 - support count is at least a few hundreds
 - in a database of several thousand transactions

Association Rule

- Association rule: an implication of the form $X \rightarrow Y$ where $X, Y \subseteq I$, and $X \cap Y = \phi$.

- $$\text{support}(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{|D|}$$
 - Transactions containing all items of *both* X and Y

- $$\text{confidence}(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)}$$

T-ID	Items
1	Bread, Ham, Juice, Cheese, Salami, Lettuce
2	Rice, Dal, Coconut, Curry leaves, Coffee, Milk, Pickle
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$R : \{\text{Bread, Salami}\} \rightarrow \{\text{Ham}\}$

$$\text{support}(R) = \frac{2}{5}$$

$$\text{confidence}(R) = \frac{2}{3}$$

Association Rule Mining Task

- Given a set of items I , a set of transactions D , a minimum support thresholds $minsup$ and a minimum confidence threshold $minconf$
- **Goal:** Find all rules R such that
 $support(R) \geq minsup$
 $confidence(R) \geq minconf$

An Example

- Transaction data

- Assume:

minsup = 30%

minconf = 80%

t1: Apple, Chicken, Milk

t2: Apple, Cheese

t3: Cheese, Boots

t4: Apple, Chicken, Cheese

t5: Apple, Chicken, Clothes, Cheese, Milk

t6: Chicken, Clothes, Milk

t7: Chicken, Milk, Clothes

- An example **frequent itemset**:

{Chicken, Clothes, Milk} [sup = 3/7]

- Association rules** from the itemset:

Clothes → Milk, Chicken [sup = 3/7, conf = 3/3]

...

...

Clothes, Chicken → Milk, [sup = 3/7, conf = 3/3]

One Approach

- Observe:

$$\text{support}(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{|D|} = \frac{\sigma(Z)}{|D|} = \text{support}(Z)$$

where $Z = X \cup Y$

- If $Z = W \cup V$, $\text{support}(X \rightarrow Y) = \text{support}(W \rightarrow V)$
 - Each binary partition of Z represents an association rule
 - With same support
 - However, the confidences may be different
- Approach: frequent itemset generation
 1. Find all itemsets Z with $\text{support}(Z) \geq \text{minsup}$. Call such itemsets *frequent itemsets*.
 2. From each Z , generate rules with $\text{confidence}(Z) \geq \text{minconf}$

Finding Frequent Itemsets

- If $|I| = n$, then number of possible itemsets = 2^n
- For each itemset, compute the support by scanning the lists of items of each transaction
 - $O(N \times w)$, where w is the average length of transactions
- Overall complexity: $O(2^n \times N \times w)$
- **Computationally very expensive!!**

Anti-monotone Property of Support

- If an itemset is *frequent*, all its subsets are also *frequent*
 - Because if $X \subseteq Y$, then $\text{support}(X) \geq \text{support}(Y)$
 - For all transactions T such that $Y \subseteq T$, we have $X \subseteq T$

$\text{Support}(\{\text{Bread}, \text{Salami}\}) \geq$
 $\text{Support}(\{\text{Bread}, \text{Ham}, \text{Salami}\})$

T-ID	Items
1	Bread, Ham, Juice, Cheese, Salami, Lettuce
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Transaction data representation

- A simplistic view of shopping baskets,
- Some important information not considered.
E.g,
 - the quantity of each item purchased and
 - the price paid.

Many mining algorithms

- There are a large number of them!!
- They use different strategies and data structures.
- Their resulting sets of rules are all the same.
 - Given a transaction data set T , and a minimum support and a minimum confidence, the set of association rules existing in T is uniquely determined.
- Any algorithm should find the same set of rules although their computational efficiencies and memory requirements may be different.
- We study only one: the Apriori Algorithm

The A-priori Algorithm

- **The best known algorithm**
- **Two steps:**
 - Find all itemsets that have minimum support (*frequent itemsets*, also called large itemsets).
 - Use frequent itemsets to **generate rules**.
- E.g., a frequent itemset
 {Chicken, Clothes, Milk} [sup = 3/7]
and one rule from the frequent itemset
 Clothes → Milk, Chicken [sup = 3/7, conf = 3/3]

The A-Priori Algorithm

Notation:

L_k = The set of frequent (large) itemsets of size k

C_k = The candidate set of frequent (large) itemsets of size.

Algorithm:

$L_1 = \{\text{Frequent 1-itemsets}\};$

for ($k = 2; L_k - 1 \neq 0; k++$) do begin

$C_k = \text{apriori_gen}(L_{k-1});$ /* Generating new candidates */

 for all transactions T in D do begin

$C_T = \text{subset}(C_k, T)$ /* Keeping only the valid candidates */

 for all candidates c in C_T do

$c.\text{count}++;$

 end

$L_k = \{c \text{ in } C_k \mid c.\text{count} \geq \text{minsup}\}$

end

Output = Union of all L_k for $k = 1, 2, \dots, n$

Generating candidate itemsets L_k

- A join of L_{k-1} with itself
insert into C_k
select $p.item_1, p.item_2, \dots, p.item_{k-1}, q.item_{k-1}$
from $L_{k-1} p, L_{k-1} q$
where $p.item_1 = q.item_1, \dots, p.item_{k-2} = q.item_{k-2},$
 $p.item_{k-1} < q.item_{k-1}$
- What does it do?

L_3	L_3
{1, 2, 3}	{1, 2, 3}
{1, 2, 4}	{1, 2, 4}
{1, 3, 4}	{1, 3, 4}
{1, 3, 5}	{1, 3, 5}
{2, 3, 4}	{2, 3, 4}

$$C_4 = \{ \{1, 2, 3, 4\}, \{1, 3, 4, 5\} \}$$

A prune step:

{1, 3, 4, 5} will be pruned
because {1, 4, 5} $\notin L_3$

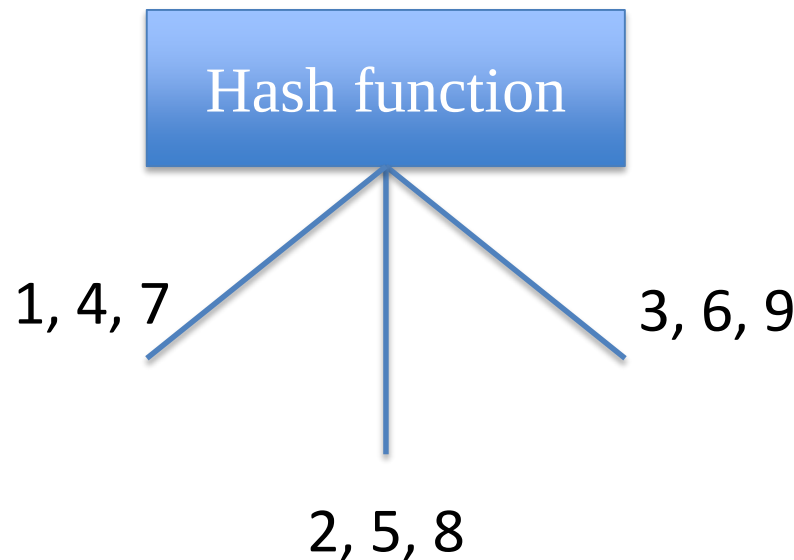
Checking Support for candidates

- One approach:
 - for each candidate itemset $c \in C_k$
 - for each transactions $T \in D$ do begin
 - check if $c \subseteq T$
 - end
 - end
- Complexity?

Using a Hash Tree

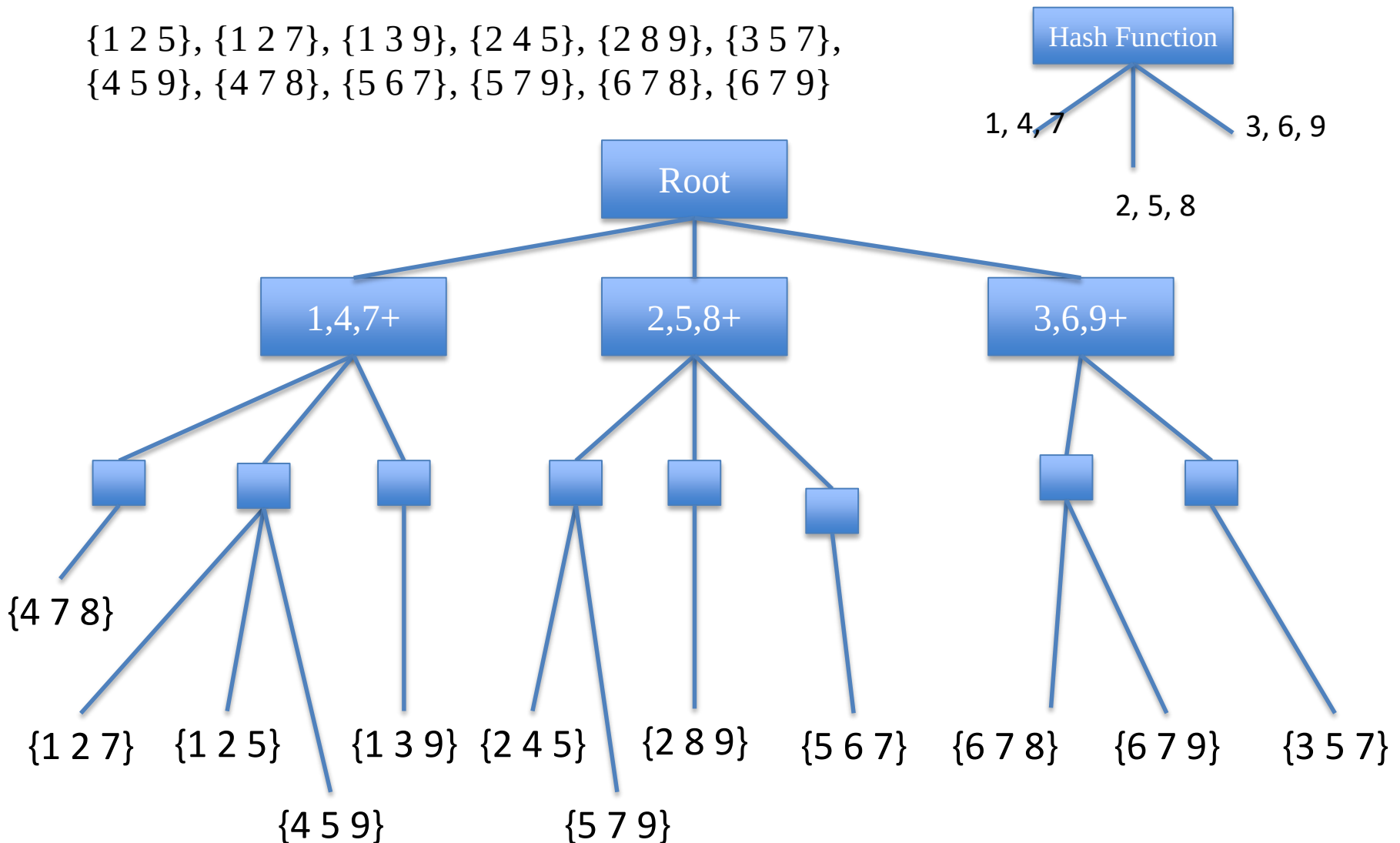
Let us have 12 candidate itemsets of size 3

{1 2 5}, {1 2 7}, {1 3 9}, {2 4 5}, {2 8 9}, {3 5 7},
{4 5 9}, {4 7 8}, {5 6 7}, {5 7 9}, {6 7 8}, {6 7 9}



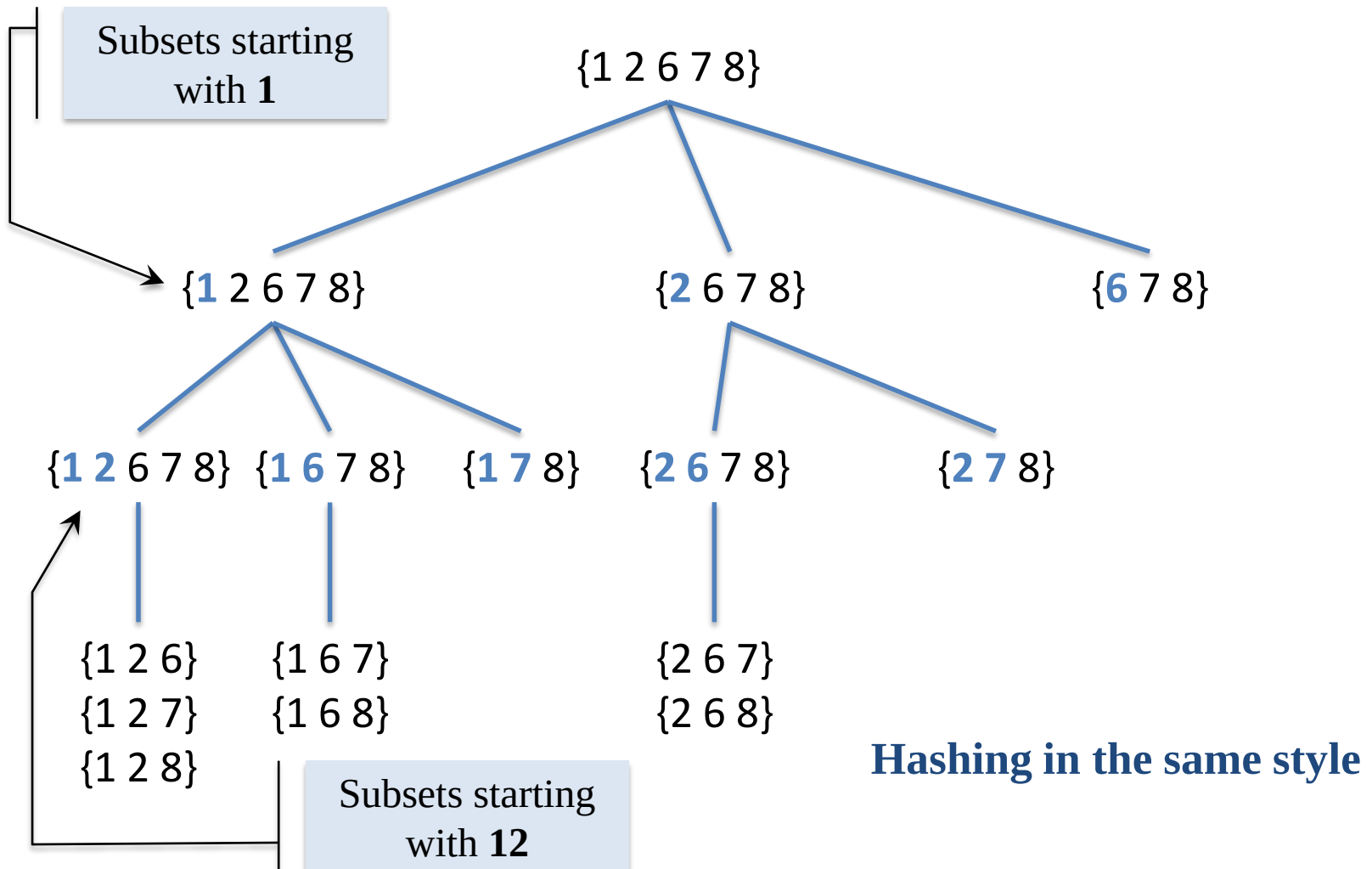
The Hash Tree

{1 2 5}, {1 2 7}, {1 3 9}, {2 4 5}, {2 8 9}, {3 5 7},
{4 5 9}, {4 7 8}, {5 6 7}, {5 7 9}, {6 7 8}, {6 7 9}



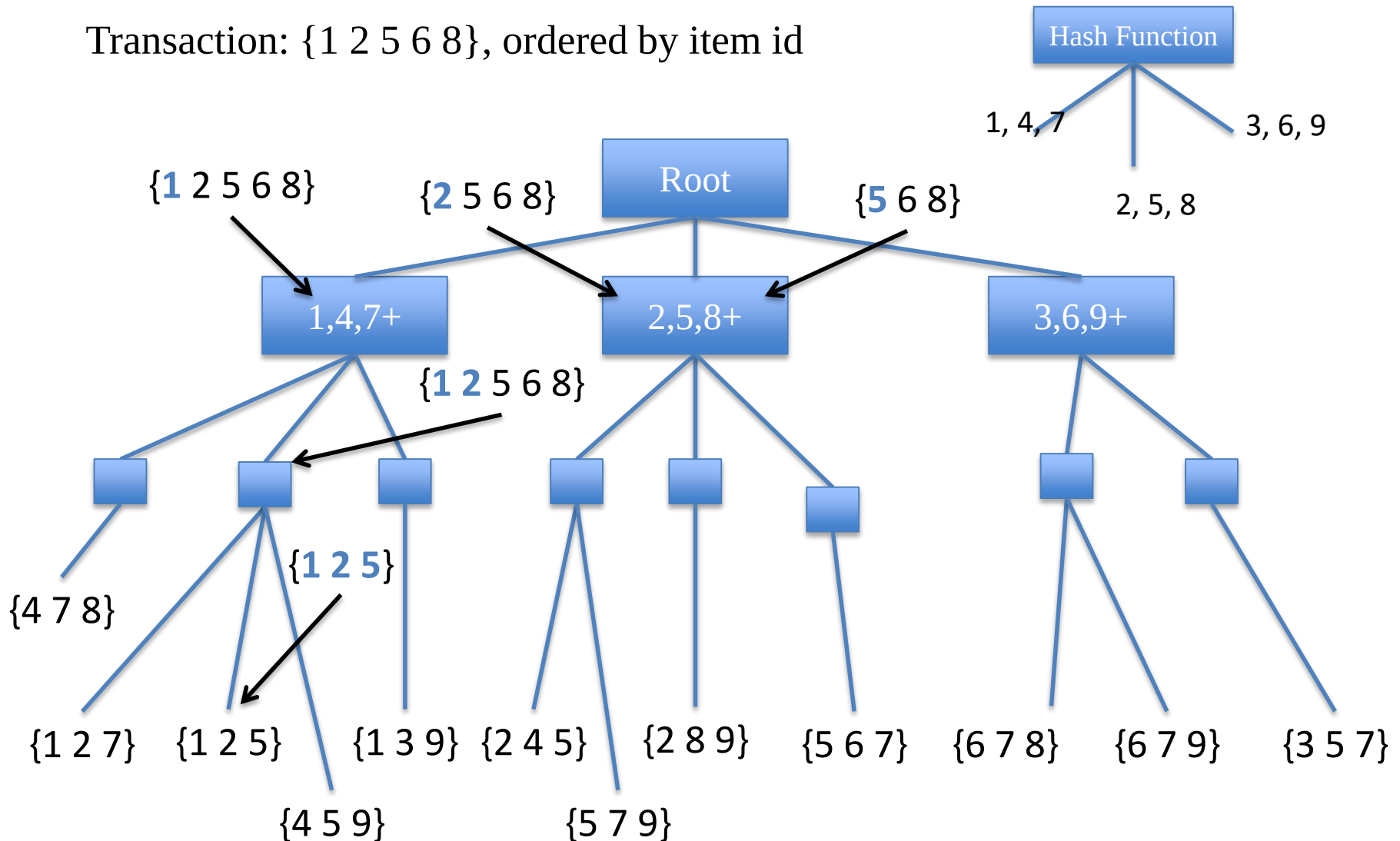
Subsets of the transaction

All subsets of size 3 for a transaction{1 2 6 7 8}, ordered by the item id



The Subset Operation using Hash Tree

Transaction: {1 2 5 6 8}, ordered by item id



Apriori: Drawbacks

- Still needed to generate a huge number of candidate sets. At each step, finding candidate set with huge number of items is a tedious task.
 - For example, if there are 10^4 frequent 1-itemsets, the Apriori algorithm will need to generate more than 10^7 candidate 2-itemsets.
- Needed to repeatedly scan the whole database and check a large set of candidates by pattern matching.
 - It is costly to go over each transaction in the database to determine the support of the candidate itemsets.
- **Solution:** Frequent Pattern growth (i.e., **FP-Growth**) using Frequent Pattern tree (i.e., **FP-Tree**) (without such a costly candidate generation process).

Frequent Pattern Growth Method (FP- growth)

- *Example:*
 - Given minimum support threshold of 3 and transactional database T:

TID	Itemset
01	f, a, c, d, g, i, m, p
02	a, b, c, f, l, m, o
03	b, f, h, j, o
04	b, c, k, s, p
05	a, f, c, e, l, p, m, n

Frequent Pattern Growth Method (FP- growth)

- First scan of database T derives a list L:

$$L = \{(f,4), (c,4), (a,3), (b,3), (m,3), (p,3)\}$$

- The root of the tree, labeled with ROOT is created.

- Scan the database T second time.

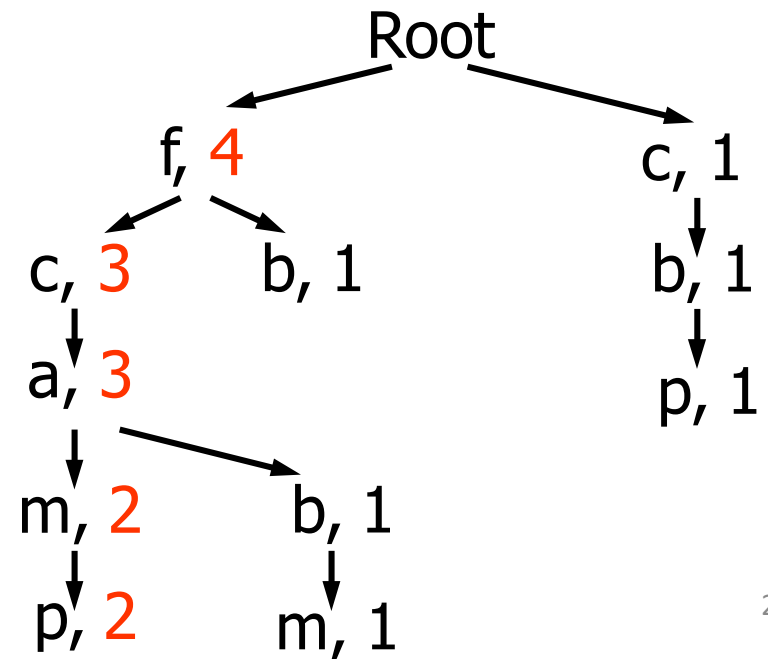
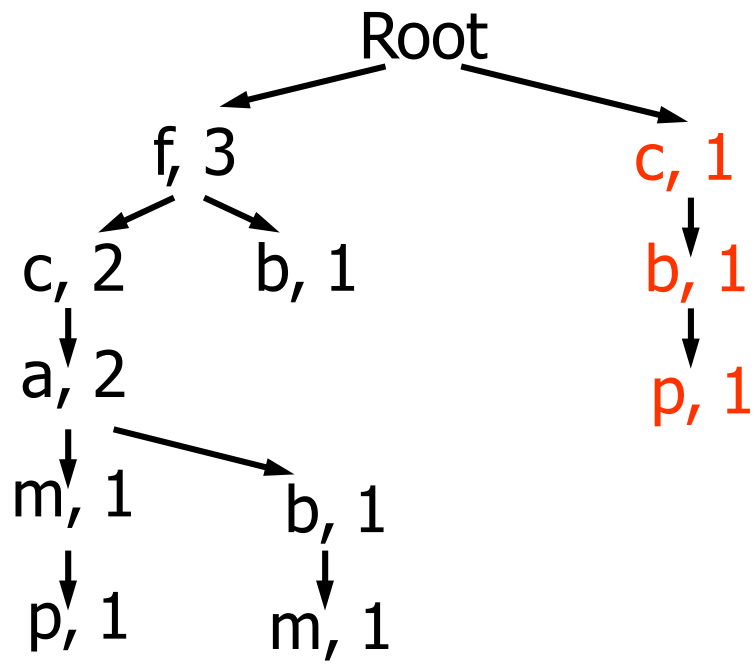
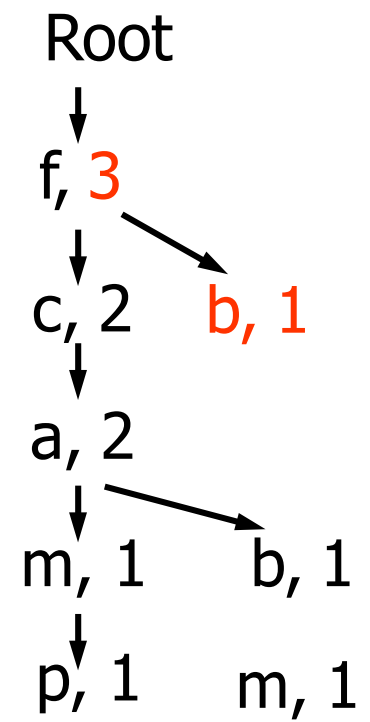
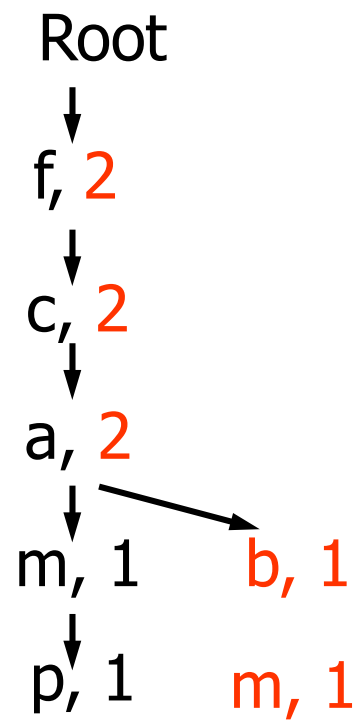
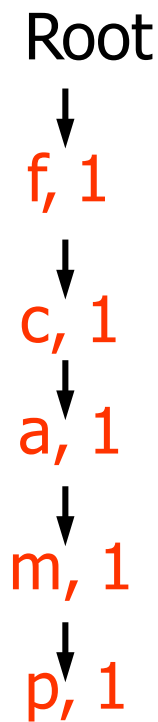
- The scan of the first transaction leads to the construction of the first branch of the FP-tree:

$$\{(f,1), (c,1), (a,1), (m,1), (p,1)\}$$

- For the second transaction: since it shares common items, f, c, and a, it shares the common prefix {f, c, a} with the previous branch, and extend to the new branch:

$$\{(f, 2), (c, 2), (a, 2), (m, 1), (p, 1)\}$$

TID	Frequent items (ordered)
01	f, c, a, m, p
02	f, c, a, b, m
03	f, b
04	c, b, p
05	f, c, a, m, p



Frequent Pattern Growth Method (FP- growth)

- To facilitate tree traversal, an item **header table** is built, in which each item in **L** list connects nodes in FP-tree with the same item value through **node- links**
- According to the list of frequent items **L**, the complete set of frequent itemsets can be divided into subsets (6 for our example) without overlap:
 1. frequent itemsets having item p (the end of L list);
 2. the itemsets having item m but no p;
 3. the frequent itemsets with b and without both m and p;
 - 4~5. ...;
 6. the large itemsets only with f.

Frequent Pattern Growth Method (FP- growth)

- For our example, two paths are selected in the FP-tree:

$\{(f,4), (c,3), (a,3), (m,2), ((p,2))\}$ and

$\{(c,1), (b,1), (p,1)\}$

- where samples with a frequent item p are

$\{(f,2), (c,2), (a,2), (m,2), ((p,2))\}$ and

$\{(c,1), (b,1), (p,1)\}$

- Given threshold value (3) satisfies only frequent itemset

$\{(c,3), (p,3)\}$, or simplified **$\{c, p\}$**

Frequent Pattern Growth Method (FP- growth)

- Next subset of frequent itemsets are these with m and without p.
 - FP-tree recognizes paths:
 $\{(f,4), (c,3), (a,3), (m,2)\}$ and
 $\{(f,4), (c,3), (a,3), (b,1), (m,1)\}$
 - corresponding accumulated samples:
 $\{(f,2), (c,2), (a,2), (m,2)\}$ and
 $\{(f,1), (c,1), (a,1), (b,1), (m,1)\}$
 - Analyzing samples we discover frequent itemset
 $\{(f,3), (c,3), (a,3), (m,3)\}$ or simplified $\{f, c, a, m\}$

Frequent Pattern Growth Method (FP- growth)

- Repeating the same process for subsets 3) to 6) in our example, additional frequent itemsets could be mined.
- In our example these are itemsets $\{f, c, a\}$ and $\{f, c\}$, but they are already subsets of the frequent itemset $\{f, c, a, m\}$.
- Therefore the final solution of the FP-growth method is the set of frequent itemsets, which is in our example:

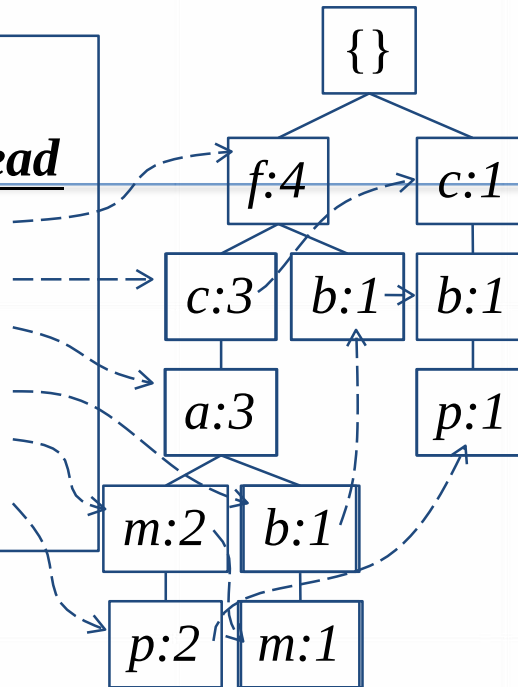
$\{ \{c, p\}, \{f, c, a, m\} \}$

- FP-growth algorithm is about an order of magnitude faster than Apriori algorithm

Header Table

Item frequency head

<i>f</i>	4
<i>c</i>	4
<i>a</i>	3
<i>b</i>	3
<i>m</i>	3
<i>p</i>	3



Conditional pattern bases

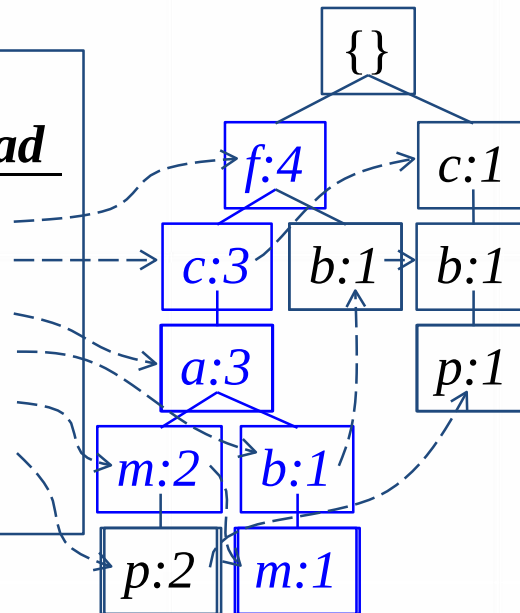
item cond. pattern base

<i>c</i>	<i>f</i> :3
<i>a</i>	<i>fc</i> :3
<i>b</i>	<i>fca</i> :1, <i>f</i> :1, <i>c</i> :1
<i>m</i>	<i>fca</i> :2, <i>fcab</i> :1
<i>p</i>	<i>fcam</i> :2, <i>cb</i> :1

Header Table

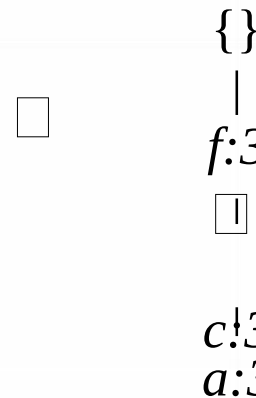
Item frequency head

<i>f</i>	4
<i>c</i>	4
<i>a</i>	3
<i>b</i>	3
<i>m</i>	3
<i>p</i>	3



m-conditional pattern base:

fca:2, *fcab*:1



All frequent patterns concerning *m*

m,
fm, *cm*, *am*,
*fc**m*, *f**a**m*, *c**a**m*,
*fca**m*

m-conditional FP-tree

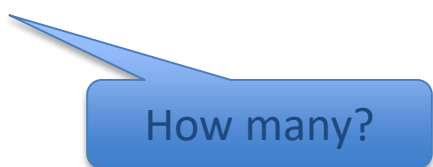
Frequent Pattern Growth Method (FP- growth)

Item	Conditional pattern-base	Conditional FP-tree
p	{(fcam:2), (cb:1)}	{(c:3)} p
m	{(fca:2), (fcab:1)}	{(f:3, c:3, a:3)} m
b	{(fca:1), (f:1), (c:1)}	Empty
a	{(fc:3)}	{(f:3, c:3)} a
c	{(f:3)}	{(f:3)} c
f	Empty	Empty

Maximal frequent itemsets: {c, p}, {f, c, a, m}

Where are we now?

- Computed *frequent itemsets*, i.e. the itemsets with required support *minsup*
- Each frequent k -itemset X gives rise to several association rules
- Ignoring $X \rightarrow \phi$ and $\phi \rightarrow X$, $2^k - 2$ rules
- Rules generated from *different* itemsets are also different
- The rules need to be checked for minimum confidence
- All these rules already satisfy the support condition



How many?

Rules Generated from the Same Itemset

- Let $X \subset Y$, for non empty itemsets X , and Y
- Then $X \rightarrow Y - X$ is an association rule
- Theorem: *If $X' \subset X$, then $c(X \rightarrow Y - X) \geq c(X' \rightarrow Y - X')$*
 - Example: $c(\{1\ 2\ 3\} \rightarrow \{4\ 5\}) \geq c(\{1\ 2\} \rightarrow \{3\ 4\ 5\})$
- *Proof.*

Observe: $c(X \rightarrow Y - X) = \sigma(Y)/\sigma(X)$

$$c(X' \rightarrow Y - X') = \sigma(Y)/\sigma(X')$$

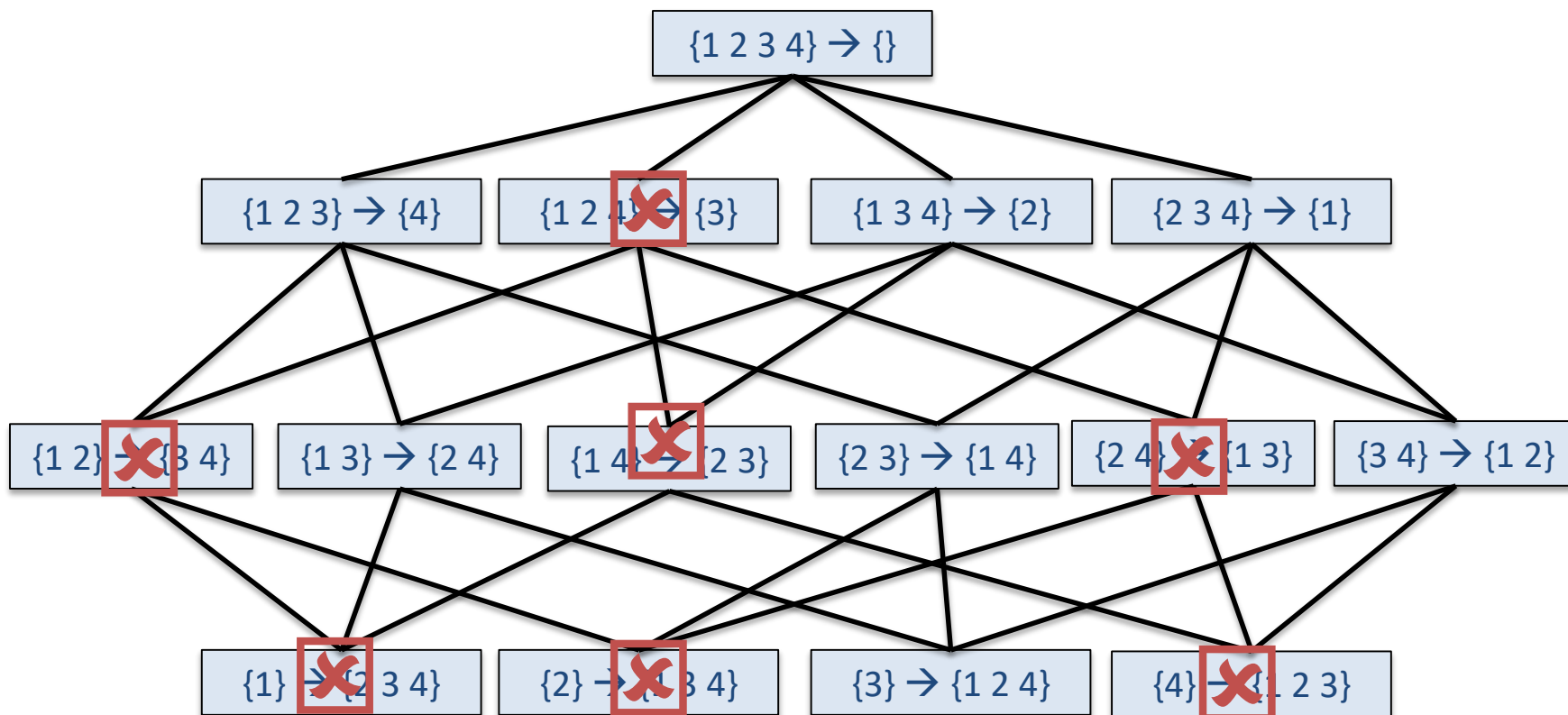
since $X' \subset X$, $\sigma(X') \geq \sigma(X)$

so $c(X \rightarrow Y - X) \geq c(X' \rightarrow Y - X')$ ■

- Corollary: *If $X \rightarrow Y - X$ is not a high-confidence association rule, then $X' \rightarrow Y - X'$ is also not a high confidence rule.*

Level-wise Approach for Rule Generation

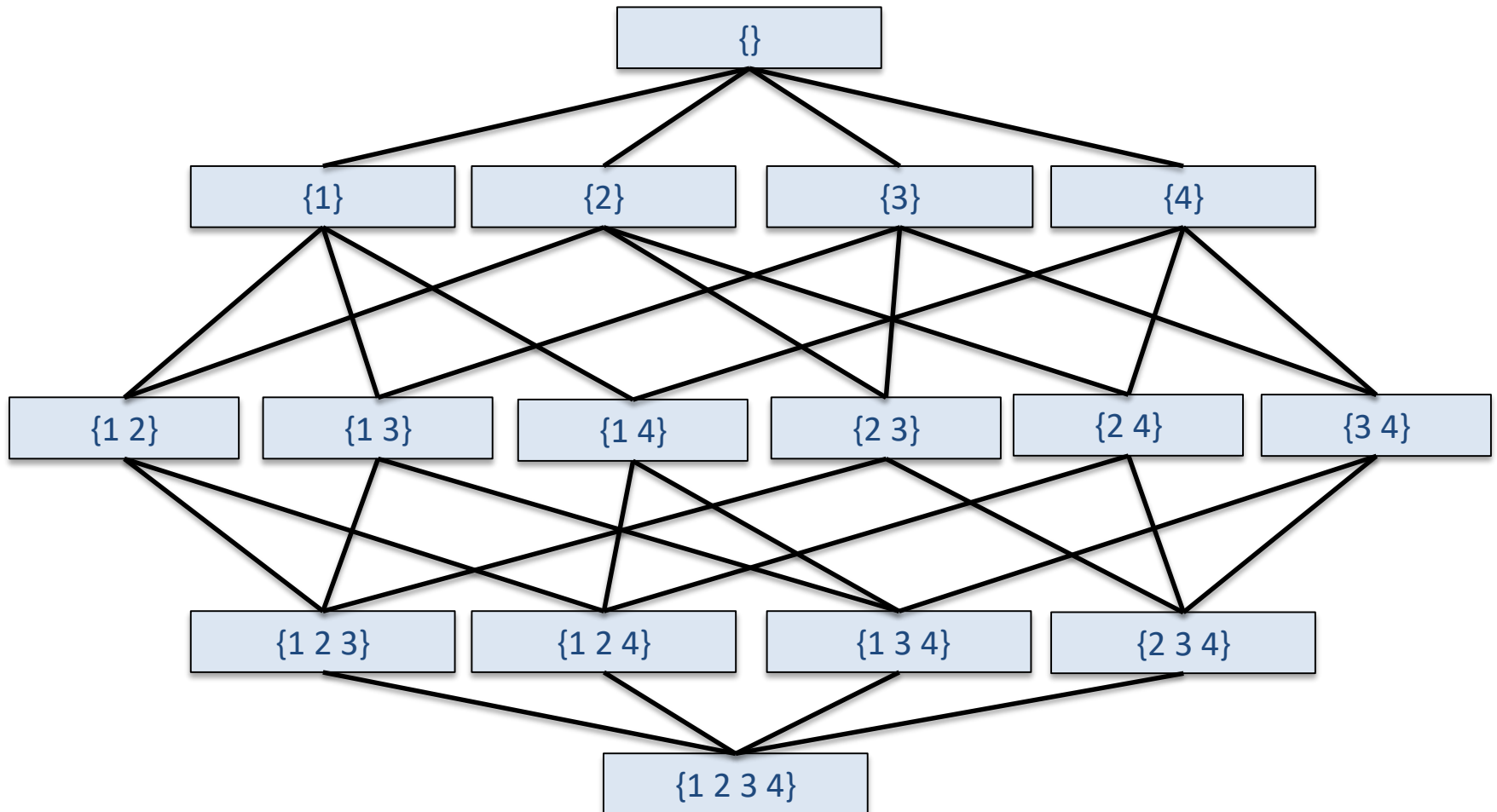
Frequent itemset: {1 2 3 4}



- Suppose $\{1\ 2\ 4\} \rightarrow \{3\}$ fails the confidence bar
- The whole tree under $\{1\ 2\ 4\} \rightarrow \{3\}$ can be discarded

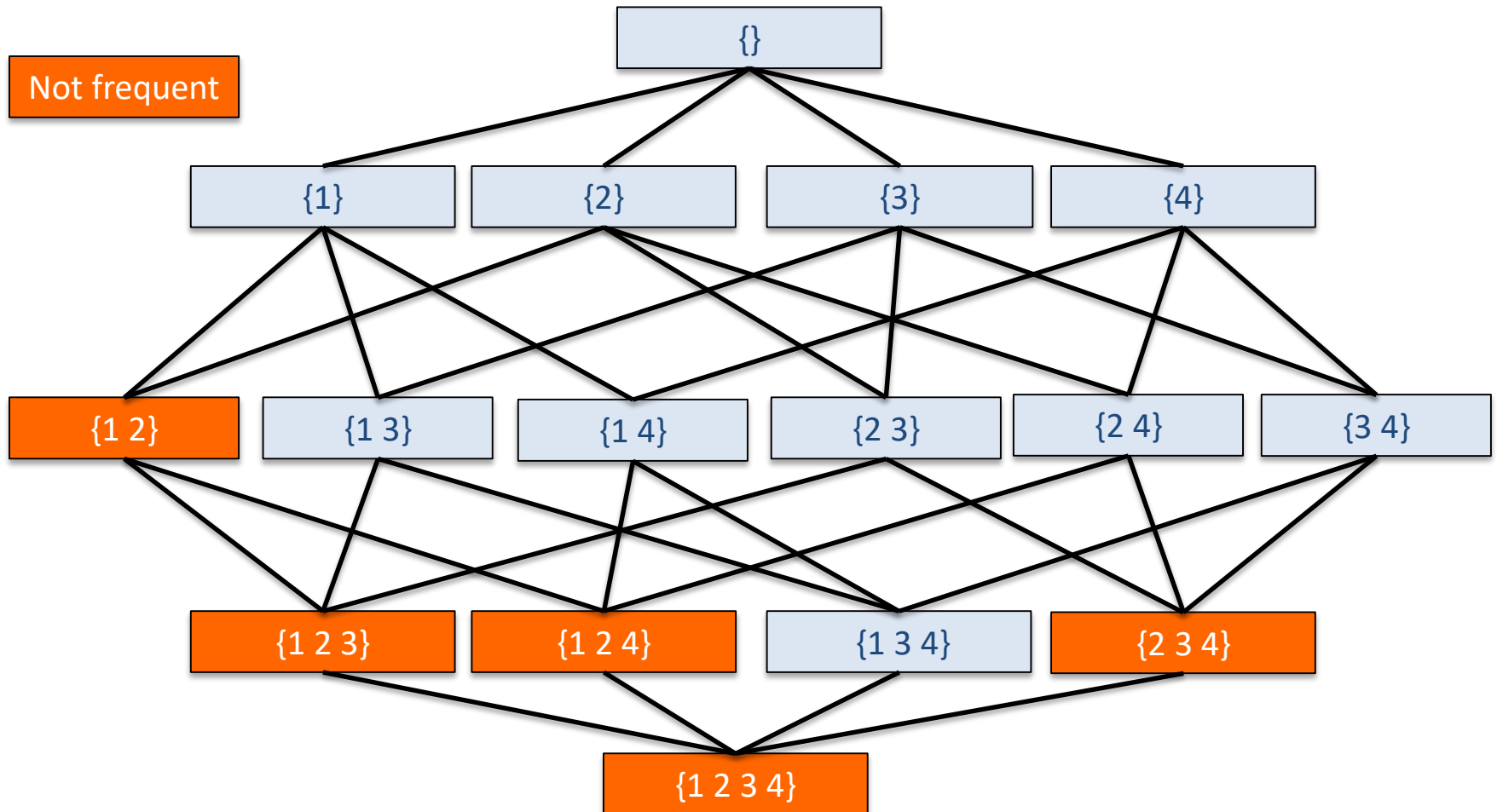
Maximal Frequent itemsets

Maximal frequent itemset: an itemset, for which none of its immediate supersets are frequent



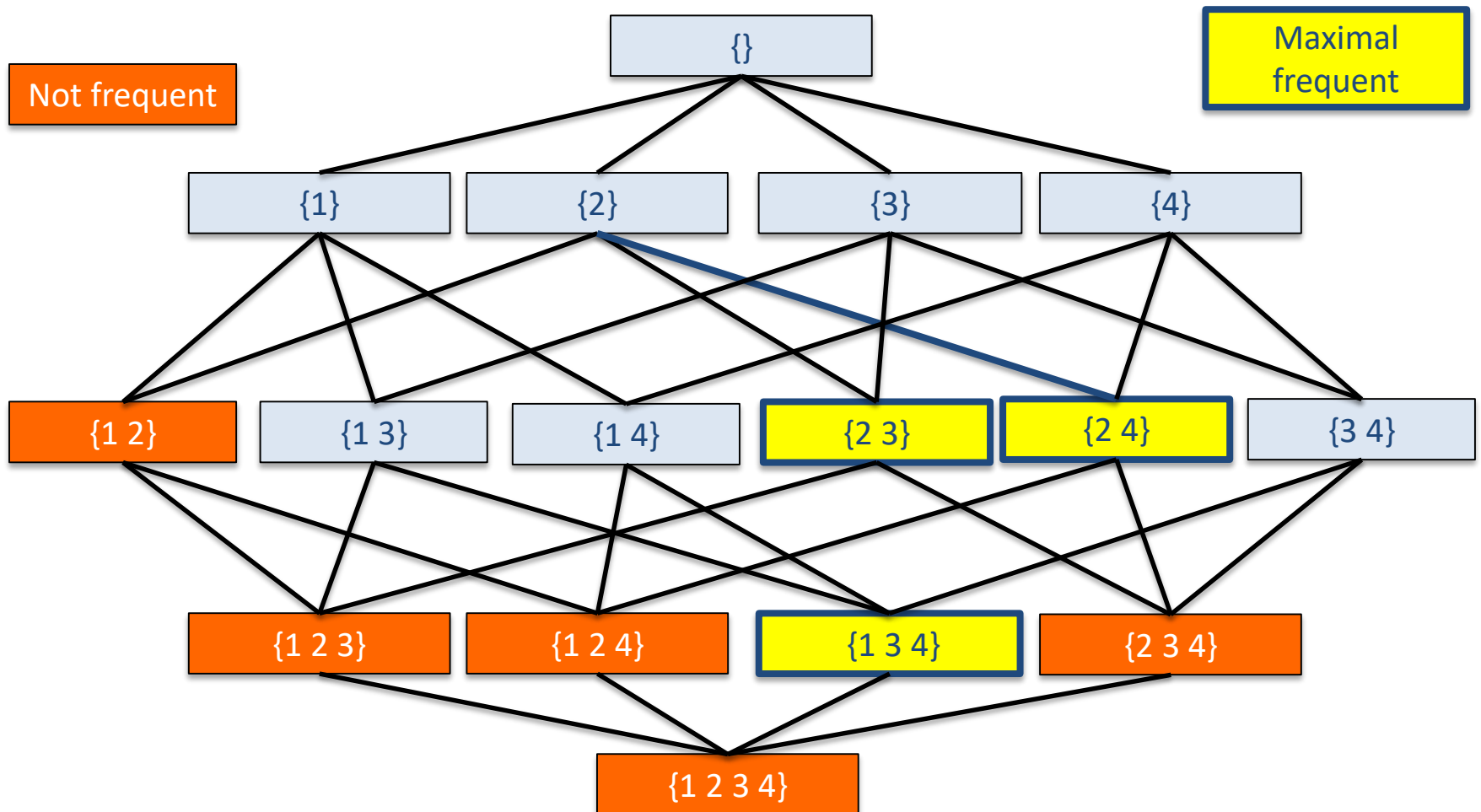
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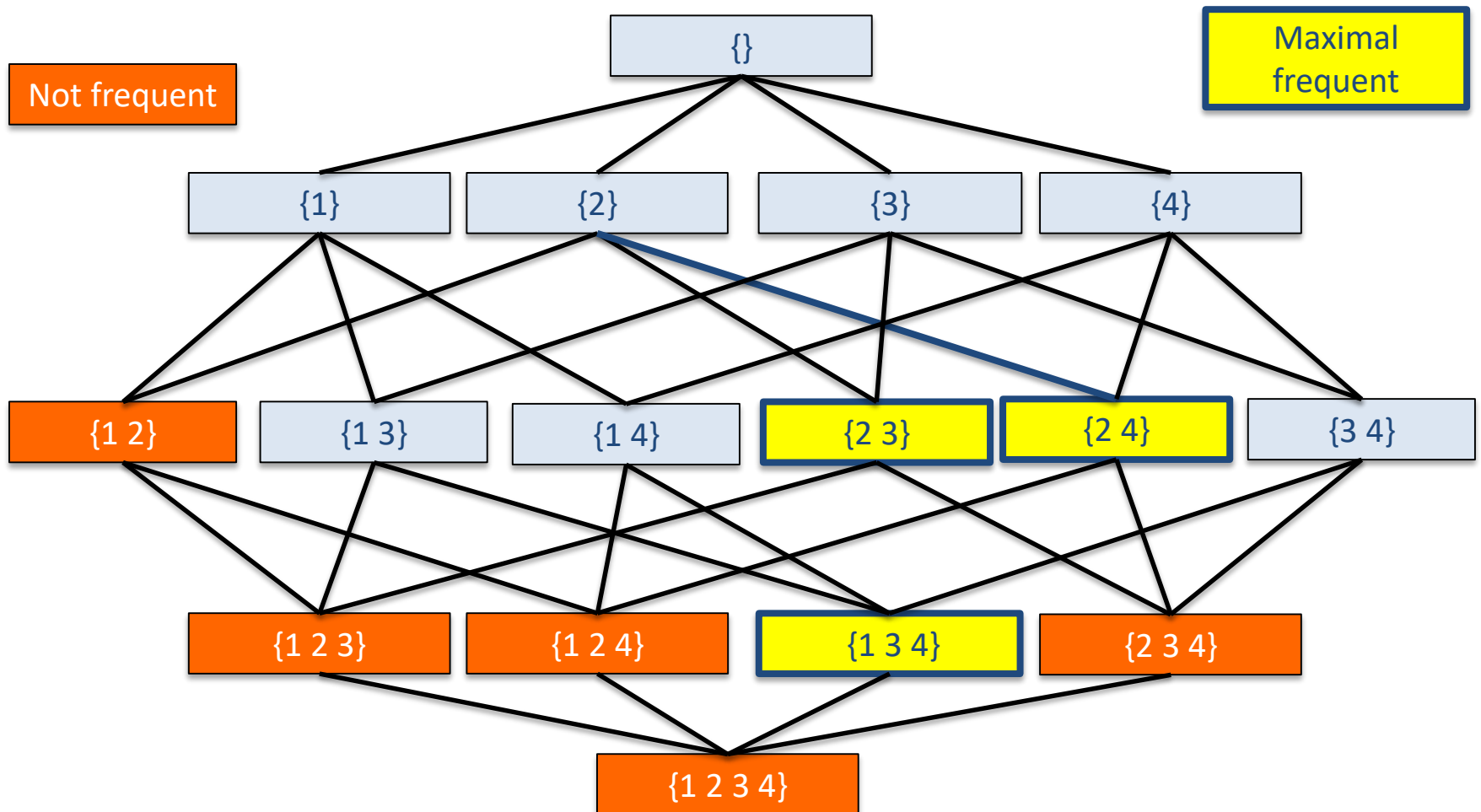
Maximal Frequent itemsets

Maximal frequent itemset: an itemset, for which none of its immediate supersets are frequent



Maximal Frequent itemsets

All frequent itemsets are subsets of one of the maximal frequent itemsets.



Maximal Frequent Itemsets

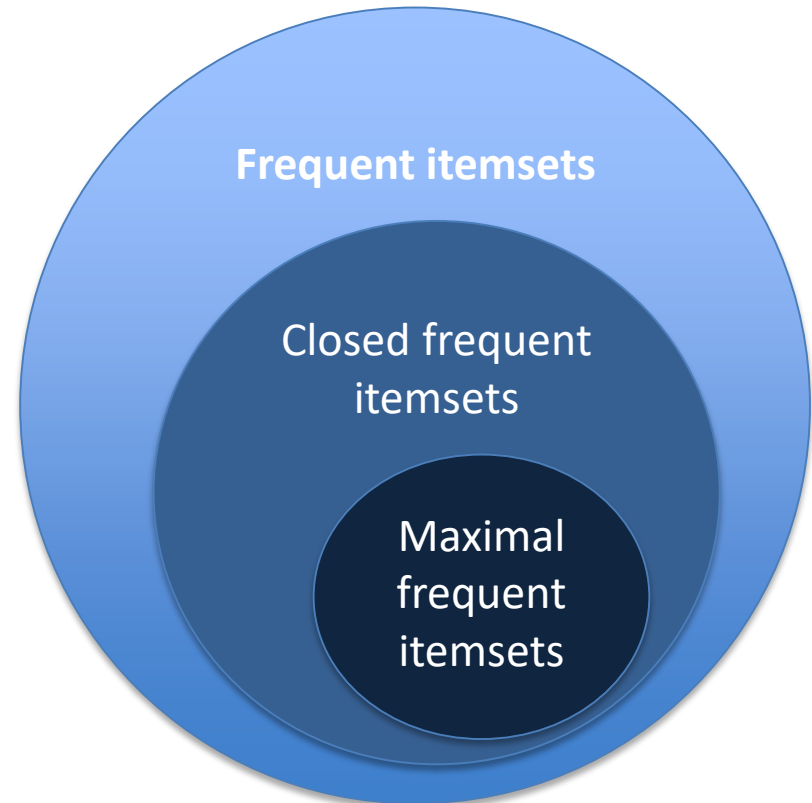
- Valuable *compact* representation of the frequent itemsets

But

- Do not contain the *support* information of the subsets
 - Says all supersets have lesser support, but does not say if any subset also has the same support

Closed Frequent Itemsets

- *Closed itemset*: an itemset X for which none of its *immediate supersets* has exactly the *same support count* as X
 - If X is not closed, at least one of its immediate supersets have the same support as the support of X
- *Closed frequent itemset*: an itemset which is *closed* and *frequent* (support $\geq \text{minsup}$)
- Support for non-closed frequent itemsets can be determined from the support information of the closed frequent itemsets



Evaluation of Association Rules

- Even from a small dataset a very large number of rules can be generated
 - For example, as support and confidence conditions are relaxed, number of rules explode
- Interestingness measure for patterns / rules is required
- Objective interestingness measure: a measure that uses statistics derived from the data
 - Support, confidence, correlation, ...
 - Domain independent
 - Requires minimal human involvement

Subjective Measure of Interestingness

- The rule $\{\text{Salami}\} \rightarrow \{\text{Bread}\}$ is *not so interesting* because it is *obvious*!
- Rules such as $\{\text{Salami}\} \rightarrow \{\text{Dish washer detergent}\}$, $\{\text{Salami}\} \rightarrow \{\text{Diper}\}$, etc are less obvious
- Subjectively more interesting for marketing experts
 - Non-trivial cross sell
- Methods for subjective measurement
 - Visualization aided: human in the loop
 - Template-based: constraints are provided for rules
 - Filter obvious and non-actionable rules

Contingency Table

	Coffee	Tea	
Tea	150	50	200
Coffee	650	150	800
	800	200	1000

	B	B'	
A	f_{11}	f_{10}	f_{1+}
A'	f_{01}	f_{00}	f_{0+}
	f_{+1}	f_{+0}	

- Frequency tabulated for a pair of binary variables
- Used as a useful evaluation and illustration tool
- Generally:

A' (or B') denotes the transactions in which A (or B) is absent

f_{1+} = support count of A

f_{+1} = support count of B

Limitations of Support & Confidence

- Tuning the support threshold is tricky
- Low threshold – Too many rules generated!
- High threshold – Potentially interesting patterns may fall below the support threshold

Limitation of Confidence

	Coffee	Coffee	
Tea	150	50	200
Tea	650	150	800
	800	200	1000

Consider the rule:

$\{\text{Tea}\} \rightarrow \{\text{Coffee}\}$

Support = 15%

Confidence = 75%

- But: Overall 80% people have coffee
 - i.e., the rule $\{\} \rightarrow \{\text{Coffee}\}$ has confidence 80%.
 - Among tea takers, the percentage actually drops to 75%!!
- Where does it go wrong?
- Confidence measure ignores the support of Y for a rule $X \rightarrow Y$

Interest factor

- *Lift*: $Lift(X \rightarrow Y) = \frac{c(X \rightarrow Y)}{\sigma(Y)}$
- For binary variables, *lift* is equivalent to *interest factor*
- *Interest factor*: $I(X, Y) = \frac{s(X \cup Y)}{s(X) s(Y)} = \frac{N f_{11}}{f_{1+} \cdot f_{+1}}$
- Similar to *baseline frequency* comparison under *statistical independence* assumption
 - If X and Y are statistically independent, their *baseline frequency* (*expected frequency of X and Y both occurring*) is

$$f_{11} = \frac{f_{1+} \cdot f_{+1}}{N}$$

Interest factor

- Intuitively

$I(X,Y) = 1$, if X and Y are independent

> 1 , if X and Y have a positive correlation

< 1 , if X and Y have a negative correlation

- Verify for the tea – coffee example

$$I = \frac{N f_{11}}{f_{1+} \cdot f_{+1}}$$

$I(\text{Tea}, \text{Coffee})$

$= 0.15 / (0.2 \times 0.8)$

$= 0.94$

	Coffee	Coffee	
Tea	150	50	200
Tea	650	150	800
	800	200	1000

Limitation of Interest Factor

	Text	Text			Mining	Mining	
Analysis	880	50	930	Graph	20	50	70
Analysis	50	20	70	Graph	50	880	930
	930	70	1000		70	930	1000

- Observe: $I(\text{Text}, \text{Analysis}) = 1.02$, $I(\text{Graph}, \text{Mining}) = 4.08$
- *Text* and *Analysis* are more related than *Graph* and *Mining*
- Confidence measure:
 - $c(\text{Text} \rightarrow \text{Analysis}) = 94.6\%$
 - $c(\text{Graph} \rightarrow \text{Mining}) = 28.6\%$
- What goes wrong here?

More Measures

- Correlation coefficient for binary variables:

$$\phi = \frac{f_{11}f_{00} - f_{01}f_{10}}{\sqrt{f_{1+}f_{+1}f_{0+}f_{+0}}}$$

- IS Measure: I and S measures combined

$$IS(A, B) = \sqrt{I(A, B) \times s(A, B)} = \frac{s(A, B)}{\sqrt{s(A)s(B)}}$$

- Mathematically equivalent to cosine measure of binary variables

Properties of Objective Measures

	B	B'	
A	f_{11}	f_{10}	f_{1+}
A'	f_{01}	f_{00}	f_{0+}
	f_{+1}	f_{+0}	

- Inversion property: Invariant under inversion operation
 - Exchange f_{11} with f_{00} and f_{01} with f_{10}
 - The value of the measure remains the same
- Null addition property: Invariant under addition of counts for *other* variables, i.e. the value of the measure remains the same if f_{00} is increased
- Which measures have which properties?

References

- Rakesh Agrawal and Ramakrishnan Srikant
Fast Algorithms for Mining Association Rules
VLDB 1994
- *Introduction to Data Mining*, by Tan, Steinbach, Kumar
 - The webpage: <http://www-users.cs.umn.edu/~kumar/dmbook/index.php>
 - Chapter 6 is available online: <http://www-users.cs.umn.edu/~kumar/dmbook/ch6.pdf>